Grade 8

Innovative Science Olympiad

2025 – 2026

SET - A

Total Questions: 50 | Time: 1.5 hr. **Total Marks: 100**

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Science

1. Why do army tanks move on continuous caterpillar tracks instead of wheels? **Select the scientific reason involving pressure.**

(A) To increase the friction between the ground and the vehicle.

(B) To decrease the total weight of the tank on the ground.

(C) To increase the surface area of contact, thereby reducing pressure on the ground.

(D) To decrease the surface area of contact, thereby increasing the grip on the ground.

2. In humans, the male reproductive system includes the testes which are located outside the abdominal cavity in a sac called the scrotum. **What is the biological significance of this anatomical arrangement?**

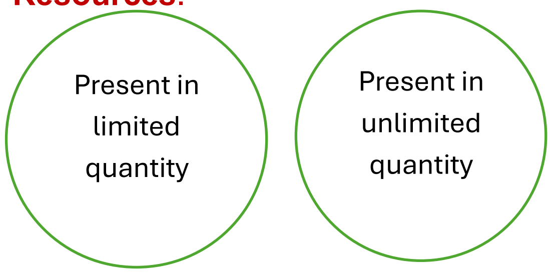
(A) To protect the testes from being crushed by internal organs.

(B) To maintain a temperature lower than the internal body temperature for sperm production.

(C) To allow the sperm to travel faster through the urethra.

(D) To ensure that the sperm can be stored for several years.

3. Two circles represent **Natural Resources.**



Circle A **Circle B**

Which of the following correctly represents the resources in Circle A and Circle B?

(A) Circle A – Air, Water; Circle B – Coal, Petroleum

(B) Circle A – Coal, Forests; Circle B – Air, Sunlight

(C) Circle A – Water, Sunlight; Circle B – Minerals, Wildlife

(D) Circle A – Air, Natural gas; Circle B – Forests, Coal

4. A door needs a minimum turning moment of 20 N·m to start opening. The handle is 40 cm away from the hinges. If a student pushes the handle at an angle of 30° with the door (not perpendicular), what is the minimum force required to open the door?

(A) 25 N (B) 50 N

(C) 100 N (D) 200 N

5. A farmer uses a **"Seed Drill"** for sowing wheat. His neighbor uses the **"Broadcasting"** (manual scattering) method. Why is the Seed Drill method more likely to result in a higher yield?

(A) It uses fewer seeds to cover a larger area.

(B) It ensures seeds are sown at uniform depths and distances, preventing overcrowding and ensuring they are covered by soil.

(C) It adds fertilizer to the seeds automatically.

(D) It kills weeds while it sows the seeds.

6. A person heats coal strongly in the absence of air. During the process, a gaseous substance is obtained which is a mixture of different hydrocarbons and carbon monoxide. It was used as a fuel in the early days and is now mainly used as a source of heat. Identify the product formed.

(A) Coke (B) Coal gas

(C) Coal tar (D) Natural gas

7. A student observes these two cases: **Situation X:** An air-filled balloon is pressed against a wall. It gets slightly compressed and stays stuck for some time.

Situation Y: A rubber suction cup is pressed on a smooth tile. It sticks tightly and can even hold some weight. The main force responsible for sticking in **both** situations is:

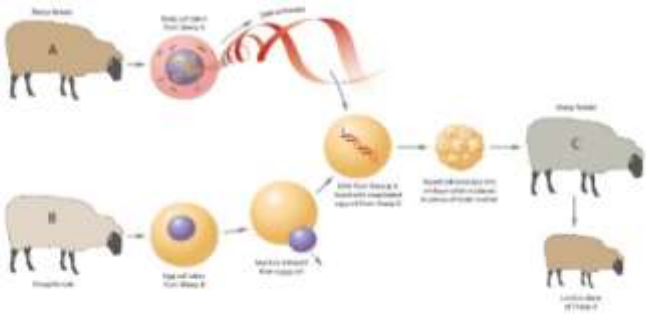
(A) Magnetic force in X and gravitational force in Y

(B) Electrostatic force in both situations

(C) Atmospheric pressure in both situations

(D) Frictional force in X and muscular force in Y

8. Cloning is the production of an exact copy of a cell or an entire organism. In the case of 'Dolly the sheep', the nucleus was taken from a mammary gland cell of a Finn Dorsett sheep and inserted into an enucleated egg of a Scottish Blackface ewe. **Why was Dolly identical to the Finn Dorsett sheep?**



(A) Because the Scottish Blackface ewe provided the nutrients.

(B) Because the mammary gland cell contained the genetic information (nucleus) of the Finn Dorsett sheep.

(C) Because the egg cell was larger than the sperm cell.

(D) Because the environment in which Dolly grew up was controlled.

9. The table below shows some petroleum products with their properties and uses. Identify the incorrectly matched row. **Which row is incorrect?**

Row	Petroleum product	State / property	Use
I	LPG	Gas under high pressure	Cooking fuel
II	Kerosene	Thin clear liquid	Fuel for lighting lamps
III	Petrol	Thin clear brown liquid	Lubrication of machine parts
IV	Bitumen	Black highly viscous liquid	Road surfacing

(A) I (B) III

(C) IV (D) II

10. A box of weight 200 N is kept on a horizontal floor. The dimensions of the box allow it to be placed such that the area of contact can be either 0.5 m² or 2 m². What is the difference in pressure exerted by the box on the floor between these two orientations?

(A) 150 Pa (B) 300 Pa

(C) 400 Pa (D) 600 Pa

11. A student is looking at a sample of stagnant pond water under a microscope. He sees a unicellular organism with "false feet" (pseudopodia) engulfing a smaller particle. **What is this organism, and what is the process called?**

(A) Yeast; Fermentation

(B) Paramecium; Ciliary movement

(C) Amoeba; Phagocytosis (or Food Vacuole formation)

(D) Spirogyra; Fragmentation

12. A city plans to replace petrol and diesel vehicles with a fuel that is less polluting, can be stored under high pressure, and is supplied through pipelines for domestic and commercial use. The fuel mainly contains methane and is also used in power generation. **Which fuel is being referred to?**

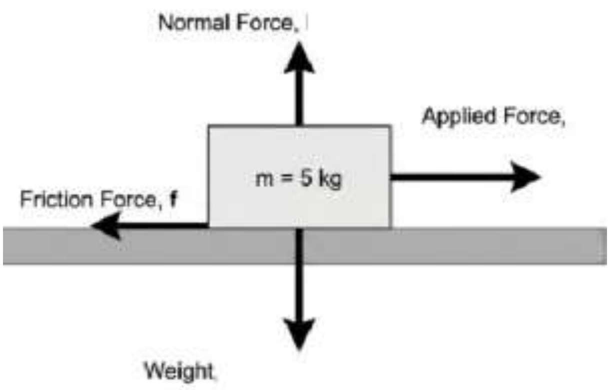
(A) LPG

(B) Coal gas

(C) Compressed natural gas (CNG)

(D) Biogas

13. A block of mass 5 kg is kept on a horizontal floor. The coefficient of static friction (μ_s)is 0.4 and the coefficient of kinetic friction (μ_k)is 0.3. If a horizontal force of 18 N is applied to the block, what will happen? *(Take $g = 10$ m/s^2)*



(A) The block will accelerate with 2 m/s^2.
(B) The block will remain at rest.
(C) The block will move with constant velocity.
(D) The block will just about to move.

14. Read the given statements and select the option that correctly identifies them as true (T) and false (F).

(i) Wood is combustible

(ii) Iron nail is non-combustible

(iii) Straw is non-combustible

(iv) Glass is combustible

	(i)	**(ii)**	**(iii)**	**(iv)**
(A)	F	F	T	F
(B)	T	T	F	F
(C)	F	T	F	T
(D)	T	T	T	T

15. A plant cell is treated with a metabolic poison that prevents the formation of "Cellulose." The cell is then placed in a solution that has a lower salt concentration than the cell's interior. **What will happen to the cell?**

(A) It will remain unchanged because the cell membrane is still intact.

(B) It will shrink because it cannot produce new organelles.

(C) It will swell and eventually burst, behaving like an animal cell.

(D) It will divide immediately to reduce the internal pressure.

16. A car of mass 1000 kg is moving on a rough road with a speed of 20 m/s. The driver applies brakes and the car comes to rest after skidding for 50 meters. What was the average frictional force offered by the road?

(A) 2000 N (B) 4000 N

(C) 8000 N (D) 500 N

17. A student sets up two "Potato Osmoscopes." In Setup A, a cavity is made in a raw potato and filled with concentrated sugar solution, then placed in a beaker of pure water. In Setup B, the potato is boiled first, then the same procedure is followed. After 2 hours, **what result will be observed?**

(A) The sugar level rises in both setups because the potato tissue is porous.

(B) The sugar level rises in Setup A but remains unchanged in Setup B.

(C) The sugar level rises in Setup B only because boiling makes the cells more absorbent.

(D) The water level in the beaker rises in both cases due to gravity.

18. Study the below reactions carefully.

$Fe + 2HCl \rightarrow FeCl_2 + H_2$

$2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$

Which of the following statements is incorrect?

(A) Both reactions are redox reactions.

(B) In both reactions, HCl acts as an oxidising agent.

(C) Iron and aluminium are reduced in their respective reactions.

(D) Hydrogen gas is formed by the reduction of H^+ ions.

19. A wooden block is placed on a plank. You slowly lift one end of the plank to create an incline. The block just begins to slide down when the plank makes an angle of 30° with the horizontal. **This angle is known as:**

(A) Angle of Friction

(B) Angle of Sliding

(C) Angle of Repose

(D) Angle of Rolling

20. An experiment is conducted on tadpoles in three separate tanks. Tank 1 has normal pond water. Tank 2 has distilled water with added Thyroxine hormone. Tank 3 has pond water where the Thyroid glands of the tadpoles have been surgically removed. **What is the most likely outcome?**

(A) Tadpoles in Tank 1 will develop faster than those in Tank 2.

(B) Tadpoles in Tank 2 will undergo rapid metamorphosis, while those in Tank 3 will remain tadpoles indefinitely.

(C) All tadpoles will die because distilled water is toxic to amphibians.

(D) Tadpoles in Tank 3 will grow into giant frogs immediately.

21. Match Column I with Column II and select the correct option.

Column I (Alloy)	Column II
P. Stainless steel	(i) Used for statues, coins and ship's propellers
Q. Brass	(ii) Alloy of copper and zinc
R. Bronze	(iii) Light and strong; used in aircraft
S. Duralumin	(iv) Hard and does not rust

(A) P→(iv), Q→ (ii), R→ (i), S→ (iii)

(B) P→ (ii), Q→ (iv), R→ (i), S→ (iii)

(C) P→ (iv), Q→ (i), R→ (ii), S→ (iii)

(D) P→ (iii), Q→ (ii), R→ (i), S→ (iv)

22. **Assertion (A):** It is dangerous to drive a car with bald (worn-out) tyres on a wet road.

Reason (R): Bald tyres have a reduced coefficient of friction with the wet road, decreasing the stopping distance.

(A) Both A and R are true, and R is the correct explanation of A.

(B) Both A and R are true, but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

23. In a hypothetical scenario, a forest is fragmented into small "islands" by the construction of several highways. Scientists find that the population of Top Carnivores (like Tigers) drops faster than the population of small rodents. **Why?**

(A) Rodents are faster and can cross highways more easily.

(B) Carnivores have a higher "home range" requirement; fragmented patches cannot provide enough prey or space to support a breeding pair.

(C) Fragmentation causes the plants to stop producing oxygen, affecting larger animals first.

(D) Rodents prefer living near highways because humans drop food there.

24. A water purification plant uses iron pipes to carry a disinfectant solution containing chlorine gas dissolved in water. Over time, the engineers notice formation of a brown solid inside the pipes and a decrease in chlorine content of the solution. **Which of the following best explains this observation?**

(A) Iron is less reactive than chlorine, so chlorine displaces iron from its oxide.

(B) Chlorine oxidises iron to iron(III) chloride, and iron reduces chlorine to chloride ions.

(C) Iron displaces chlorine from chloride ions because iron is more reactive than chlorine.

(D) Chlorine reacts only with water and not with iron, so rust forms independently.

25. A thundercloud is at a distance of 3.3 kmfrom an observer. If the speed of sound in air is 330 m/sand the speed of light is 3×10^8 m/s, what is the time gap between seeing the lightning and hearing the thunder?

(A) 0.1 seconds (B) 10 seconds

(C) 100 seconds (D) 1 second

26. A scientist is testing the efficacy of a new antibiotic. He grows bacteria on a Petri dish and places a disc soaked in the antibiotic in the center. After 24 hours, he sees a **"Zone of Inhibition"** of 15 mm. If he repeats the experiment with a **"Resistant"** strain of the same bacteria, what will happen?

(A) The Zone of Inhibition will be significantly smaller or non-existent.

(B) The Zone of Inhibition will become 30 mm because resistant bacteria are more sensitive.

(C) The bacteria will turn into a fungus to survive.

(D) The antibiotic will evaporate faster because of the resistance.

27. **Assertion:** Gold can remain in the free state in nature and does not get easily corroded by air or water.
Reason: Gold has very little tendency to react and dissolves only in aqua regia.
Choose the correct option.

(A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(C) Assertion is true, but Reason is false.

(D) Assertion is false, but Reason is true.

28. Match the following terms in **Column-I** with their correct defining characteristics in **Column-II.**

Column I	Column II
(P) Ultrasound	(i) Used to distinguish sounds of same pitch & loudness
(Q) Quality/Timbre	(ii) Frequency below 20 Hz
(R) Infrasound	(iii) Frequency above 20,000 Hz
(S) Noise	(iv) Irregular and non-periodic vibrations

(A) P-iii, Q-i, R-ii, S-iv

(B) P-ii, Q-i, R-iii, S-iv

(C) P-iii, Q-iv, R-ii, S-i

(D) P-i, Q-iii, R-iv, S-ii

29. Assertion (A): Reforestation is the same as Afforestation.

Reason (R): Reforestation involves planting trees in an area where a forest previously existed but was destroyed

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.

(B) Both Assertion (A) and Reason (R) are true, but R is not the correct explanation of A.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

30. Four situations are given below:

I. Coal burning in a furnace with a sufficient supply of air

II. Burning of firecrackers

III. Burning of coal in a closed room with limited air supply

IV. Sodium metal exposed to air and catching fire without external heating

Which of the following statements is correct regarding these situations?

(A) Situation II represents spontaneous combustion and Situation IV represents explosion.

(B) Situation I represents complete combustion and Situation III represents incomplete combustion.

(C) Situation III represents complete combustion because carbon dioxide is formed.

(D) Situation IV represents incomplete combustion because oxygen is not required.

31. A boy shouts "HELLO" near a cliff and hears an echo. To hear a clear echo, the reflected sound must come back at least 0.1 s after the original sound. If the speed of sound is 340 m/s, **what is the minimum distance the boy should be from the cliff?**

(A) 34 m (B) 17 m

(C) 10 m (D) 340 m

32. Assertion (A): A cell placed in a highly concentrated salt solution will undergo plasmolysis.

Reason (R): The cell membrane is selectively permeable, allowing water to move from a higher concentration inside the cell to a lower concentration outside.

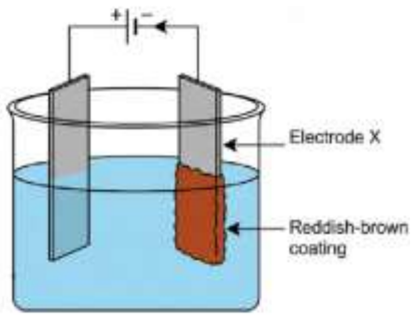
(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.

(B) Both Assertion (A) and Reason (R) are true, but R is not the correct explanation of A.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

33. Two copper plates are dipped in a blue solution and connected to a battery. After passing electric current for some time, a reddish-brown coating is observed on electrode **X**.



Identify the metal deposited on electrode **X**, the electrolyte used, and the name of electrode **X** respectively.

(A) Copper, copper sulphate solution, cathode

(B) Copper, copper sulphate solution, anode

(C) Zinc, zinc sulphate solution, cathode

(D) Copper, zinc sulphate solution, anode

34. A person stands between two parallel cliffs and fires a gun. He hears the first echo after **2 seconds** and the second echo after **3 seconds**. If the speed of sound is **340 m/s, what is the distance between the two cliffs?**

(A) 340 m (B) 850 m

(C) 1700 m (D) 510 m

35. At a summer picnic, a bowl of potato curry was left in the sun for several hours. By evening, it developed a foul smell and a slimy texture. **Which of the following is responsible for this change?**

(A) The potato starch reacted with sunlight to produce oxygen gas.

(B) Microorganisms produced toxic substances as they multiplied in the warm food.

(C) The salt in the curry evaporated, leaving behind a bitter residue.

(D) Nitrogen from the air turned the spices into a liquid state

36. When an electric current is passed through acidulated water in an electrolysis apparatus, hydrogen and oxygen gases are obtained at the electrodes.

This process is an example of which type of chemical reaction?

(A) Combination reaction

(B) Decomposition reaction

(C) Displacement reaction

(D) Neutralisation reaction

37. Two plane mirrors are placed at an angle of 60° to each other. A ray of light strikes the first mirror parallel to the second mirror. After reflection from the first mirror, it strikes the second mirror. What will be the angle of reflection at the second mirror?

(A) 30° (B) 60°

(C) 90° (D) 0°

38. Consider the life cycle of a "Butterfly" and a "Human." Which of the following is a major difference in their developmental process?

(A) The human undergoes metamorphosis, while the butterfly does not.

(B) The butterfly undergoes metamorphosis (larva to pupa to adult), while the human shows direct development.

(C) The butterfly has internal fertilization, while the human has external.

(D) The human hatches from an egg, while the butterfly is born alive.

39. A student places strips of four different metals separately into aqueous solutions of iron(II) sulphate. The following observations are made:

Metal P shows no reaction.
Metal Q displaces iron from the solution.
Metal R reacts slowly and produces hydrogen gas with dilute acid but does not displace iron.
Metal S displaces hydrogen from dilute hydrochloric acid but does not react with iron(II) sulphate.
Which of the following represents the correct decreasing order of reactivity of the metals P, Q, R and S?

(A) Q > S > R > P (B) Q > R > S > P

(C) S > Q > R > P (D) R > S > Q > P

40. You read a book under sunlight. The black printed letters appear black because they:

(A) Reflect all colours of light that fall on them.

(B) Absorb all colours of light that fall on them.

(C) Refract all light away from your eyes.

(D) Emit black light.

41. When we say a forest is a **"Carbon Sink,"** what do we mean in terms of the environment?

(A) The forest floor is full of coal and carbon deposits.

(B) The trees absorb Carbon Dioxide from the atmosphere for photosynthesis, helping to reduce the greenhouse effect.

(C) The forest releases Carbon into the soil to make it black.

(D) The forest prevents the Carbon in the air from turning into Oxygen.

42. When a lot of dry powder of substance X is released over a fire, the fire gets extinguished. Name another substance that behaves like X.

(A) Sodium carbonate

(B) Sodium hydroxide

(C) Potassium bicarbonate

(D) Potassium Chloride

43. A boy is running towards a large mirror in a dance studio at a speed of **5 m/s**. At the same moment, the mirror is mounted on a trolley and starts moving away from him at a speed of **2 m/s**. How fast will the boy see his image moving relative to him?

(A) 3 m/s (B) 7 m/s

(C) 10 m/s (D) 6 m/s

44. Why is it recommended to sow seeds at a "Uniform Distance" from each other using a seed drill, rather than just scattering them?

(A) It makes the field look more beautiful from a distance.

(B) It ensures that each plant gets sufficient sunlight, nutrients, and water without competing with its neighbor.

(C) It prevents the wind from blowing the plants away.

(D) It allows the farmer to walk through the field without touching the plants.

45. A researcher is studying a specific species of salamander. He finds that the female lays her eggs in a moist burrow, but fertilization occurs only after the male deposits a packet of sperm (spermatophore) that the female later picks up with her body. How would you classify this reproductive strategy?

(A) Purely external fertilization, because the eggs are laid in the burrow.

(B) Internal fertilization, because the sperm enters the female's body before meeting the egg.

(C) Asexual reproduction, because there is no direct physical contact between the parents.

(D) Budding, because the eggs are kept in a burrow.

46. An object is placed exactly in the middle of two mirrors joined at **60°**. A student removes a small piece of the mirror surface at the exact point where an image should form. **What happens to that specific image?**

(A) The image disappears completely.

(B) The image shifts to the left.

(C) The image is still visible because it is virtual and formed by rays from other parts of the mirror.

(D) Multiple smaller images are formed.

47. Now, read the given passage and fill in the blanks by selecting an appropriate option.

In respiration, glucose combines with **(i)** ______ to form carbon dioxide and water with the release of energy. Oxygen combines with haemoglobin to form **(ii)** ______, which supplies oxygen for the **(iii)** ______ of food. Compared to combustion, respiration is a **(iv)** ______ process.

	(i)	(ii)	(iii)	(iv)
(A)	Nitrogen	Carboxyhaemoglobin	Reduction	Fast
(B)	Oxygen	Oxyhaemoglobin	Oxidation	Slow
(C)	Oxygen	Oxyhaemoglobin	Reduction	Fast
(D)	Carbon dioxide	Carbaminohaemoglobin	Oxidation	Slow

48. During the "Life Cycle of a Silk Moth," the transition from the 'Caterpillar' stage to the 'Pupa' stage involves the spinning of a cocoon. **What is the chemical nature of the thread produced, and what is its biological purpose for the moth?**

(A) It is made of carbohydrates to provide food for the pupa.

(B) It is a protein fiber that hardens on exposure to air to protect the pupa during its transformation into an adult.

(C) It is a waxy substance used to keep the pupa waterproof.

(D) It is a calcium-based shell similar to a bird's egg.

49. farmer uses an electric pump fitted with a metal probe to check whether water in his irrigation tank is suitable for running the pump safely. When the probe is dipped into rainwater, the indicator bulb does not glow. However, when the same probe is dipped into well water, the bulb glows brightly.
Which of the following conclusions is most appropriate?

(A) Rainwater does not contain ions, whereas well water contains dissolved salts that produce ions.

(B) Rainwater is acidic and therefore cannot conduct electricity.

(C) Well water conducts electricity because it contains free electrons like metals.

(D) Both rainwater and well water conduct electricity equally, but the bulb is faulty.

50. A technician is studying the preservation of fruit juices. He notices that juice in Bottle A, treated with Sodium Metabisulfite, remains fresh, while Bottle B, which was only boiled, spoils after three days. **What is the chemical role of Sodium Metabisulfite here?**

(A) It acts as a flavoring agent that hides the smell of decay.

(B) It is a chemical preservative that inhibits the growth of bacteria and molds by creating an environment hostile to microbial enzymes.

(C) It provides extra sugar for the yeast to grow faster and consume all the oxygen.

(D) It turns the juice into a solid form that bacteria cannot eat.

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